

RibbonTok: Multi-Resolution, Single-Stream, and Adaptive-Length Tokenization for Autoregressive Image Generation

Anonymous submission

Abstract

Autoregressive (AR) image generation relies on visual tokenizers whose sequence length, order, and information allocation determine causal cost. Fixed 2D grids couple token count to raster size; flexible 1D sequences reduce this cost but need not preserve discrete identity when the same logical content appears on different grids. We present RibbonTok, a resolution-consistent discrete 1D representation that separates logical description from rendering grid. Every externally selected prefix conditions reconstruction, while multi-resolution coalesce matching aligns common-prefix assignment distributions across logically equivalent views. A masked-register ViT and four directional codebooks produce discrete positions; a rectified-flow DiT decodes any prefix on a supported grid. On ImageNet-256, RibbonTok-XL reaches rFID 1.04 at 256 positions, and 32-position class-conditional generation with a 3B causal backbone obtains gFID 1.43. The complete system is more than $2\times$ faster end to end than a conservative LlamaGen-3B comparison proxy.

Introduction

Autoregressive (AR) image generation recasts synthesis as next-token prediction over a learned visual vocabulary (Yu et al. 2024a; Sun et al. 2024; Wang et al. 2026). This interface reuses causal transformers, classifier-free guidance, and KV caching, but makes sequence length, ordering, and information allocation determine both the AR learning problem and its sequential cost.

Most discrete visual tokenizers quantize a two-dimensional latent grid (van den Oord, Vinyals, and Kavukcuoglu 2017; Esser, Rombach, and Ommer 2021; Luo et al. 2024). Rasterization yields a long, locally correlated 1D sequence with arbitrary scan order and canvas-dependent length. Compact 1D tokenizers reduce this redundancy (Yu et al. 2024b), and flexible variants make several prefix lengths decodable (Yan et al. 2025; Duggal et al. 2025; Miwa et al. 2025; Bachmann et al. 2025). However, three properties should be separated. *Prefix decodability* asks whether an early subsequence is useful; *grid flexibility* asks whether a decoder can render different spatial sizes; and *representation consistency* asks whether shared logical content keeps the same discrete identity after its physical raster changes. A variable-length or variable-grid decoder does not by itself provide the third property.

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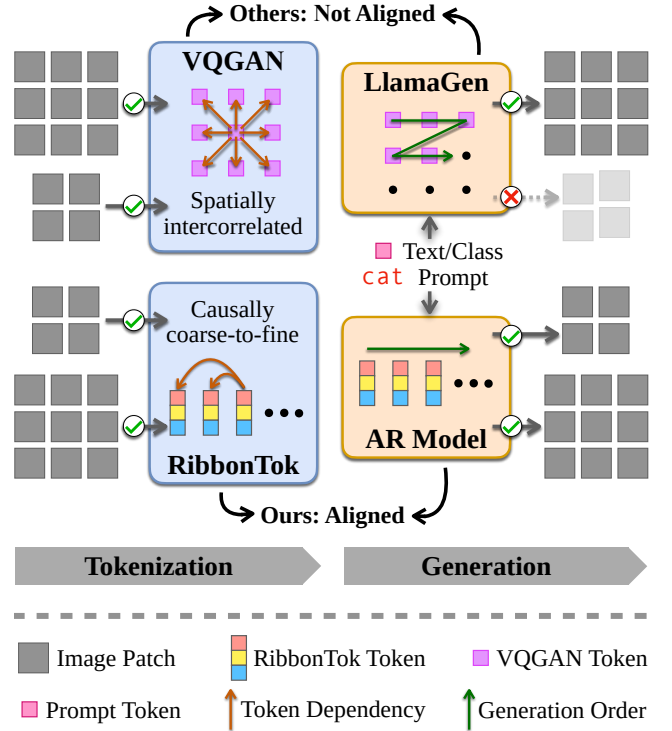


Figure 1: Representation axes in RibbonTok. A fixed 2D tokenizer couples token count, ordering, and output canvas. RibbonTok uses an ordered 1D description: prefix length controls retained evidence, while the target grid controls physical resolution. Multi-resolution training aligns common prefixes across rasters. Short prefixes condition plausible reconstructions and do not uniquely specify every pixel.

We introduce RibbonTok, which treats a token sequence as a *logical image description* and the target latent grid as a separate *physical rendering request* (Figure 1). One image is represented by a single coarse-to-fine stream, and every prefix can condition reconstruction. Leading positions capture global layout and semantics; trailing positions progressively reduce ambiguity and constrain finer appearance (Figure 2). The prefix length is supplied by the user or a downstream policy, so the present system offers explicit rate-quality con-

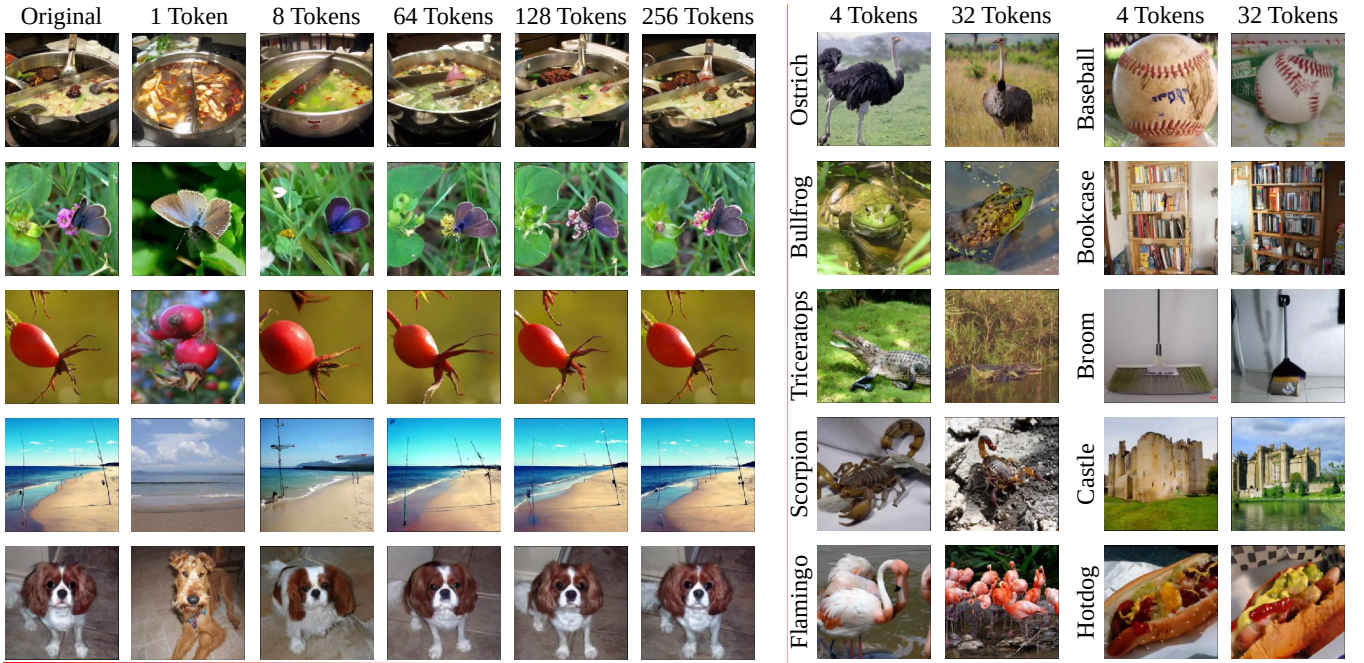


Figure 2: Coarse-to-fine reconstruction and short-sequence generation. *Left*: nested prefixes provide progressively stronger evidence to the detokenizer. *Right*: class-conditioned samples use 32 AR positions with four parallel indices each. The budget is selected externally, and short prefixes leave fine appearance for the generative detokenizer to complete.

tol rather than learned per-image stopping. Crucially, the same generated prefix can condition any supported target grid without re-tokenization or a resolution-specific AR sequence.

Figure 3 summarizes the architecture. A fixed VAE maps pixels to latent patches. A ViT with causally masked learned registers (Darcet et al. 2024) resamples those patches into an ordered sequence. Each visual position is quantized by four directional codebooks, providing high combinatorial capacity through moderate-size tables. A rectified-flow DiT cross-attends to any quantized prefix and generates a requested VAE latent grid. Because this detokenizer is generative, a short prefix specifies a conditional distribution of plausible reconstructions rather than preserving every pixel; longer prefixes supply progressively more image-specific evidence.

To train token identity across raster changes, we distinguish *physical resolution* (the observed or rendered grid) from *logical resolution* (the recoverable detail). For each training image, we form the original view, a downsampled view, and a nearest-neighbor re-upsampled copy. The last two have different physical sizes but the same logical detail and token length. Our *coalesce matching* operation gives all three views one assignment distribution over their common logical prefix, while trailing-token dropping teaches every prefix to remain useful (Figure 4). This objective targets a stronger notion than decoder resize support: common logical content is trained toward a shared discrete prefix, and that prefix can then be rendered on different supported grids.

The resulting representation directly supports causal generation (Figure 5). One AR step predicts four categorical

indices in parallel, and generation may stop at any selected prefix length. This design shifts much of image formation from a long sequential AR loop to a fixed-step flow detokenizer whose spatial computation is parallel. Although detokenization is not free, it carries work that would otherwise require hundreds of causal predictions; end-to-end latency is therefore the relevant efficiency measure.

On ImageNet-256, RibbonTok-XL reaches rFID 1.39 with 32 visual positions and 1.04 with 256. With 1B and 3B causal backbones, 32-position generation obtains gFID 1.68 and 1.43. The 3B system requires 0.65 s for AR sampling and 2.87 s end to end on one A100, versus a conservative > 5.98 -s LlamaGen-3B proxy formed from our 256-step AR lower bound plus its reported detokenization cost; it also improves source-reported gFID from 2.18 to 1.43 and IS from 263.3 to 348.8. Native-grid reconstruction and ablations isolate the effects of causal ordering, prefix training, stochastic assignment, and cross-resolution coalescing.

Contributions.

- We introduce a prefix-decodable discrete 1D representation that factorizes an image’s logical description from its physical rendering grid, allowing one prefix to condition multiple supported output resolutions.
- We propose multi-resolution triplet training with coalesce matching, which explicitly aligns the common-prefix assignment distribution of views that share logical content across different physical rasters.
- We demonstrate strong reconstruction and 32-position AR generation, together with native-resolution, ablation,

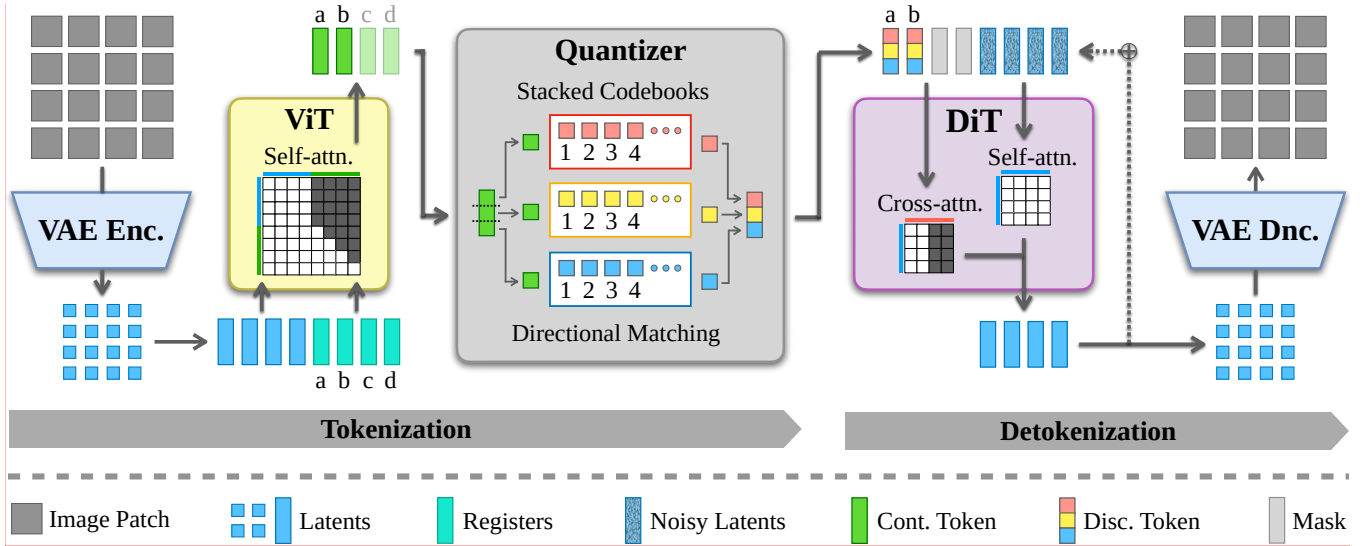


Figure 3: RibbonTok inference. *Tokenization*: a fixed VAE maps pixels to latent patches; a masked-register ViT resamples them into a causally ordered 1D stream; and a stacked directional-codebook quantizer maps each visual position to four discrete indices. *Detokenization*: any prefix cross-conditions a rectified-flow DiT, which generates a latent grid at a requested supported resolution before VAE decoding. Prefix length and output-grid size are independent controls, so the same discrete prefix can be rendered on multiple supported physical grids.

and end-to-end efficiency analyses that account for both sequential generation and generative detokenization.

Related Work

Compact, flexible, and resolution-general tokenization.

TiTok replaces a spatial grid with a compact 1D register sequence (Yu et al. 2024b). ElasticTok, ALIT, CAT, and One-D-Piece vary capacity through nested masking, recurrent allocation, external complexity estimation, or quality-controllable prefixes (Yan et al. 2025; Duggal et al. 2025; Shen et al. 2025; Miwa et al. 2025). FlexTok is the closest prior work in prefix structure: it combines causal registers, nested dropout, finite scalar quantization, and a rectified-flow decoder to produce ordered 1–256-token sequences (Bachmann et al. 2025). ReToK studies information allocation within such sequences, AdaTok learns a content-conditioned stopping policy, ChannelTok orders channel-wise tokens for lightweight flexible decoding, and concurrent CaTok binds causal intervals to a MeanFlow decoder (Fu et al. 2026; Lu et al. 2026; Paul, Bansal, and Goldstein 2026; Chen et al. 2026). VibeToken is the closest resolution-general system: it decouples dynamic 1D token length from input and output raster size and supports broad resolutions and aspect ratios (Patel et al. 2026). RibbonTok addresses a complementary representation-level question not implied by output-grid support alone: when rasterization changes but logical detail is shared, should the common prefix retain its discrete identity? Coalesce matching trains that common-prefix assignment directly, separating assignment-level resolution consistency from decoder-side resolution generality.

Autoregressive generation and generative detokenizers. Discrete tokens support masked and causal image genera-

tors (Chang et al. 2022; Weber et al. 2024; Sun et al. 2024), while MAR uses continuous tokens with a per-token diffusion objective (Li et al. 2024). Latent diffusion, diffusion transformers, rectified flow, and flow matching provide high-capacity generative decoders for compact conditions (Rom-bach et al. 2022; Peebles and Xie 2023; Liu, Gong, and Liu 2023; Lipman et al. 2023); DiTo likewise studies diffusion as a scalable tokenizer objective (Chen et al. 2025). Such detokenizers suit ambiguous short prefixes but may synthesize details not explicitly encoded by them. We therefore distinguish paired reconstruction from plausible completion and report detokenization cost together with AR sequence cost.

Method

RibbonTok separates two controls that are usually entangled: prefix length determines the amount of discrete *logical evidence*, while the requested latent grid determines the *physical rendering resolution*. We seek a single 1D stream whose order supports next-token prediction, whose every prefix can condition reconstruction, and whose common logical prefix is stable across raster changes. Figures 3, 4, and 5 summarize inference, training, and generation.

RibbonTok Inference

Latent interface and token budget. For an RGB image $\mathbf{x} \in \mathbb{R}^{H \times W \times 3}$, a fixed VAE encoder produces a latent grid $\mathbf{z} = E_{\text{vae}}(\mathbf{x})$, which is patchified into N embeddings. The default number of output positions scales with the input area,

$$L = \max\left(1, \left\lfloor \frac{HW}{H_{\max}W_{\max}} L_{\max} \right\rfloor\right), \quad (1)$$

where $L_{\max} = 256$ and $H_{\max} \times W_{\max}$ is the largest training raster. Any sequence can be truncated to a prefix of

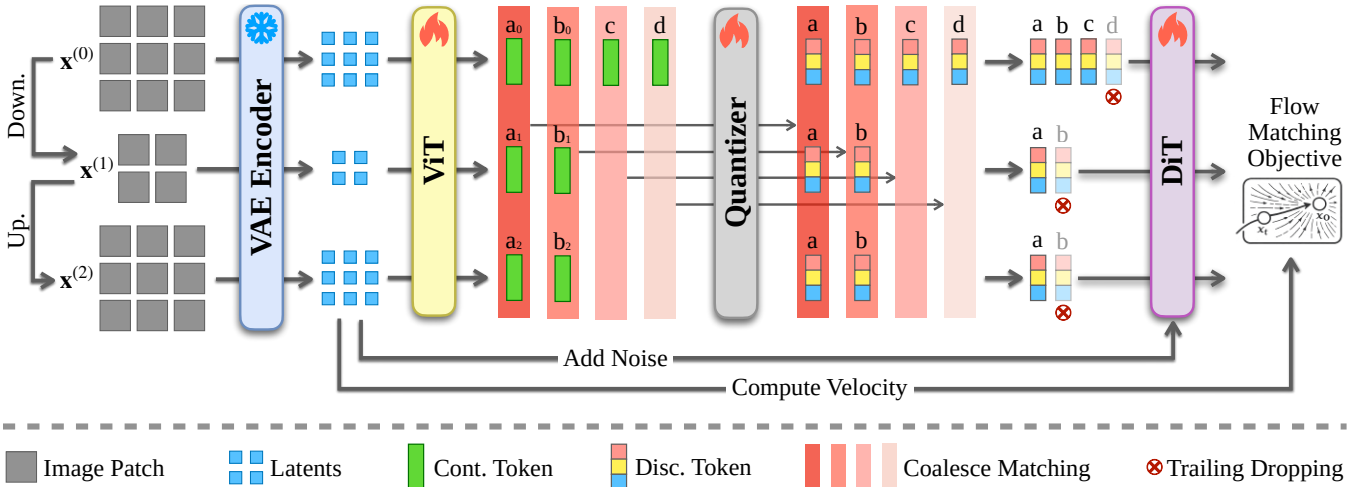


Figure 4: Multi-resolution RibbonTok training. A full-resolution image $\mathbf{x}^{(0)}$, a downsampled view $\mathbf{x}^{(1)}$, and its nearest-neighbor re-upscaled view $\mathbf{x}^{(2)}$ form a triplet. Views $\mathbf{x}^{(1)}$ and $\mathbf{x}^{(2)}$ share logical resolution and token length despite different physical rasters. *Coalesce matching* averages their common-prefix assignment logits with those of $\mathbf{x}^{(0)}$, yielding one shared assignment distribution (and identical deterministic tuples under coalesced argmax). *Trailing-token dropping* randomly retains only a prefix for flow matching, training every supported budget. All modules except the VAE are optimized jointly.

length $q \leq L$. Equation (1) is a default operating rule, not a learned stopping policy. At detokenization, q and the supported output-grid size are selected independently.

Masked-register ViT. We prepend L learned registers to the latent embeddings and process the joint sequence with a ViT. Latent queries attend only to latent keys. Register query ℓ attends to all latent keys and registers $1:\ell$, but not to future registers. The outputs $\mathbf{u}_{1:L}$ therefore form a causal dependency order: changing a later register cannot alter an earlier position. The registers retain the resampling role of register-based vision transformers (Darcet et al. 2024), while the mask turns them into an explicit coarse-to-fine stream.

Stacked directional-codebook quantizer. Each $\mathbf{u}_\ell \in \mathbb{R}^d$ is split into K chunks. For unit code vector $\mathbf{e}_j^{[k]}$, the directional assignment logit is

$$s_{\ell j}^{[k]} = \left\langle \frac{\mathbf{u}_\ell^{[k]}}{\|\mathbf{u}_\ell^{[k]}\|_2}, \mathbf{e}_j^{[k]} \right\rangle, \quad \|\mathbf{e}_j^{[k]}\|_2 = 1. \quad (2)$$

We use $K = 4$, $V = 16,384$, and $d = 256$. Thus one temporal visual position is the tuple $\mathbf{y}_\ell = (y_\ell^{[1]}, \dots, y_\ell^{[K]}) \in \{1, \dots, V\}^K$. The factorization supplies a large combinatorial symbol space while retaining four moderate lookup tables and four parallel AR heads. At evaluation, each chunk is assigned to its maximum-similarity entry. During training, we sample from the temperature-scaled similarity logits and pass the selected vector through a straight-through estimator. Exponential moving averages update the code vectors. A commitment penalty pulls each encoder chunk toward its selected stop-gradient code, while an entropy regularizer favors both decisive per-token assignments and broad aggregate code use.

Rectified-flow detokenizer. A DiT receives a noisy VAE latent grid and cross-attends to any quantized prefix $\tilde{\mathbf{u}}_{1:q}$. Timestep conditioning follows DiT (Peebles and Xie 2023); 2D rotary embeddings encode grid positions (Su et al. 2024). For a clean latent $\mathbf{z}_0$, Gaussian noise ϵ , and $t \sim \mathcal{U}(0, 1)$, rectified-flow matching (Liu, Gong, and Liu 2023; Lipman et al. 2023) uses $\mathbf{z}_t = (1 - t)\mathbf{z}_0 + t\epsilon$ and trains the predicted velocity toward $\epsilon - \mathbf{z}_0$. We use 25 integration steps by default (Esser et al. 2024). Cross-attention makes prefix length independent of latent-grid size: the same $\mathbf{y}_{1:q}$ can condition each supported target grid. This is rendering flexibility, not a claim that a short prefix uniquely fixes all pixels; later tokens progressively reduce the conditional uncertainty.

RibbonTok Training

Physical and logical resolution. Physical resolution is the observed raster size; logical resolution is the recoverable detail. Given a full-resolution image $\mathbf{x}^{(0)}$, we sample a lower square resolution and construct

$$\mathbf{x}^{(1)} = \text{Down}(\mathbf{x}^{(0)}), \quad \mathbf{x}^{(2)} = \text{Up}_{\text{NN}}(\mathbf{x}^{(1)}). \quad (3)$$

Views $\mathbf{x}^{(0)}$ and $\mathbf{x}^{(2)}$ share physical size, whereas $\mathbf{x}^{(1)}$ and $\mathbf{x}^{(2)}$ share logical detail. During triplet training, $\mathbf{x}^{(2)}$ is assigned the same length as $\mathbf{x}^{(1)}$, isolating logical content from the larger raster on which it is displayed.

Coalesce matching. For shared position $\ell \leq L^{(1)} = L^{(2)}$, codebook k , and entry j , let $s_{\ell j}^{(i)[k]}$ be the assignment logit for view i . Before stochastic quantization, we replace the three logits by

$$\bar{s}_{\ell j}^{[k]} = \frac{1}{3} \sum_{i=0}^2 s_{\ell j}^{(i)[k]}. \quad (4)$$

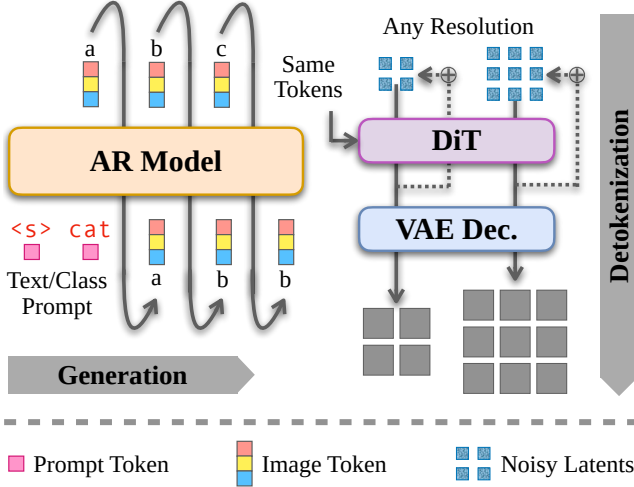


Figure 5: Autoregressive generation with RibbonTok. A class-conditioned causal transformer predicts one visual position per step; four V -way heads produce that position’s code indices in parallel. Sampling stops at an externally chosen length, after which the resulting logical prefix conditions the same rectified-flow detokenizer at any supported target grid. Thus 32 visual positions require 32 causal steps, not 128.

The triplet then shares one assignment distribution over its common logical prefix; under coalesced deterministic argmax, the corresponding code tuples are identical. Full-resolution positions after $L^{(1)}$ remain view-specific. Coalesce matching thus trains token identity across physical raster changes rather than merely enabling a decoder to resize its output. Because coalescing precedes stochastic sampling, it aligns the categorical distribution itself rather than adding a post hoc feature-distance penalty after quantization.

Trailing-token dropping and joint objective. For each view, we sample a retained length uniformly from 1 to its available length and condition the DiT only on that prefix. This trailing-token-dropping objective (Rippel, Gelbart, and Adams 2014; Bachmann et al. 2025; Miwa et al. 2025) forces early positions to support reconstruction and lets later positions specialize in refinement. During training, assignments are sampled from temperature-scaled logits to reduce early collapse. For sampled code $y_\ell^{(i)[k]}$, the encoder commitment loss is

$$\mathcal{L}_{\text{commit}} = \sum_{i,k,\ell} \left\| \mathbf{u}_\ell^{(i)[k]} - \text{sg} \left(\mathbf{e}_{y_\ell^{(i)[k]}}^{[k]} \right) \right\|_2^2, \quad (5)$$

where sg denotes stop-gradient. Let $\mathbf{p}_m^{[k]}$ index the M assignment distributions for codebook k in a minibatch. We use

$$\mathcal{L}_{\text{entropy}} = \sum_{k=1}^K \left[\frac{1}{M} \sum_{m=1}^M H(\mathbf{p}_m^{[k]}) - H \left(\frac{1}{M} \sum_{m=1}^M \mathbf{p}_m^{[k]} \right) \right], \quad (6)$$

where $H(\mathbf{p}) = -\sum_j p_j \log p_j$. The first term favors decisive assignments, and the negative aggregate term favors broad marginal code use. The trainable tokenizer and detokenizer modules minimize

$$\mathcal{L} = \frac{1}{3} \sum_{i=0}^2 \mathcal{L}_{\text{fm}}^{(i)} + \lambda_c \mathcal{L}_{\text{commit}} + \lambda_e \mathcal{L}_{\text{entropy}}. \quad (7)$$

The codebooks use the exponential-moving-average updates described above, and the VAE remains fixed.

RibbonTok for Image Generation

The four indices at each position are embedded independently and concatenated into the causal-transformer width. Four independent heads predict the next tuple. For condition $\mathbf{c}$, training minimizes

$$\mathcal{L}_{\text{AR}} = - \sum_{\ell=1}^L \sum_{k=1}^K \log p \left(y_\ell^{[k]} \mid \mathbf{c}, \mathbf{y}_{1:\ell-1} \right). \quad (8)$$

Class-conditioning dropout enables classifier-free guidance, and standard KV caching applies. At inference, a selected stopping length controls both sequential AR computation and how much discrete evidence reaches the detokenizer. The current model supports multiple budgets but does not infer an image-dependent budget itself.

Experiments

Implementation Details

Data, metrics, and models. We train on ImageNet-1K and evaluate on all 50K validation images (Russakovsky et al. 2015), at 256×256 unless noted. ImageNet is a standard large-scale class-conditional benchmark, enabling direct comparison with prior image tokenizers and generators. Reconstruction uses rFID (Heusel et al. 2017), PSNR, and SSIM; generation uses gFID and Inception Score (IS) (Salimans et al. 2016). Cross-paper FID values in Table 1 are source-reported, so the table compares complete systems rather than isolating tokenizer quality under one codebase. The fixed VAE has 16 channels and $8 \times$ spatial downsampling; we sample its posterior during training, use its mode at evaluation, and patchify its scaled latents with patch size 2. All RibbonTok variants use $L_{\text{max}} = 256$, $d = 256$, and four 16K directional codebooks. M/L/XL pair ViTs of 12×768 , 18×1152 , and 18×1152 with DiTs of 12×768 , 18×1152 , and 28×1792 . The resulting tokenizer–detokenizer systems have 296M, 949M, and 2.541B parameters, respectively.

Images are resized, randomly cropped at the maximum target resolution, and horizontally flipped. We uniformly sample a lower logical resolution from the training schedule, obtain the low-resolution view by area downsampling, and form its matched-raster view by nearest-neighbor upsampling. We train for 20K steps with a global batch size of 3,840. AdamW uses $\beta = (0.9, 0.99)$, weight decay 10^{-3} , peak learning rate 3.84×10^{-4} , 1K linear warmup steps, cosine decay, and gradient clipping at 3. Assignment temperature decreases linearly from 0.2 to 0.05 over the first 4K steps; codebook EMA decay is 0.995. The commitment and entropy weights are 0.25 and 0.05, and straight-through residual dropout is 0.2.

Tokenizer	Rep.	Vocab.	Rec.	Prefix	ID	rFID↓	PSNR↑	SSIM↑	Generator	Size	Gen.	gFID↓	IS↑
SD-VAE (2022)	2D cont.	—	1024	No	No	0.74	25.68	0.820	DiT-XL (2023)	675M	—	2.27	278.2
MAR-VAE (2024)	2D cont.	—	256	No	No	1.22	—	—	MAR-H (2024)	943M	256	1.55	303.7
MaskBit (2024)	2D disc.	16K	256	No	No	1.37	21.50	0.56	MaskBit (2024)	305M	256	1.52	328.6
TiTOK-S (2024b)	1D disc.	4K	128	No	No	1.71	17.80	—	MaskGIT-L (2022)	287M	128	1.97	281.8
OM2 (2024)	2D disc.	262K	256	No	No	1.17	22.64	—	OM2-XL (2024)	1.5B	256	2.33	271.8
LlamaGen [†] (2024)	2D disc.	16K	576	No	No	0.94	—	—	LGen-3B (2024)	3.1B	576	2.18	263.3
FlexTok (2025)	1D disc.	64K	32	Yes	No	1.08	17.70	0.489	GPT AR (2025)	1.33B	32	1.71	285.0
VibeToken-LL (2026)	1D disc.	4K×8	256	Yes	No	0.40	25.04	0.819	VibeToken-Gen	1.5B	64	3.62	226.9
ChannelTok [‡] (2026)	1D disc.	2 ¹⁶	256	Yes	No	3.70	—	—	—	—	—	—	—
AdaTok (2026)	1D disc.	—	256	Yes	No	1.31	—	—	—	—	—	—	—
RibbonTok-M	1D disc.	16K×4	256	Yes	Yes	3.82	15.42	0.465	—	—	—	—	—
RibbonTok-L	1D disc.	16K×4	256	Yes	Yes	1.27	17.85	0.530	—	—	—	—	—
RibbonTok-XL	1D disc.	16K×4	256	Yes	Yes	1.04	19.36	0.582	Llama AR	1B	32	1.68	293.7
									Llama AR	3B	32	1.43	348.8

Table 1: ImageNet reconstruction (left) and class-conditioned generation (right). “Rec.” and “Gen.” are source-reported sequence-length budgets; one sequence position may contain multiple parallel indices, so bit rates and token semantics differ by method. “Prefix” denotes multiple decodable prefixes. “ID” is intentionally narrower than output-resolution support: it denotes an explicit objective that aligns discrete common-prefix assignments across views with shared logical content and different physical rasters. Values are not controlled for training data, architecture, guidance, or sampling. OM2 and LGen abbreviate Open-MAGVIT2 and LlamaGen. VibeToken is included because it is the closest resolution-general 1D system; its excellent full-budget reconstruction and 64-position generation are reported under mixed-resolution training. [†]LlamaGen generates at 384×384 with 576 tokens and resizes for 256×256 evaluation. [‡]ChannelTok is a June 2026 arXiv preprint; its 256-token result uses channel slices as tokens and BSQ, making it a contemporaneous flexible-length reference rather than a controlled head-to-head comparison. RibbonTok reconstruction uses 25 detokenization steps; generation uses 32 positions, top- $p = 0.95$, and CFG 3.0.

Autoregressive setup and counting convention. The generators are 1B- and 3B-parameter Llama-style causal transformers (Meta AI 2024), trained without image augmentation in three 10K-step stages with maximum lengths 64, 128, and 256, a global batch size of 1,280, and class-conditioning dropout of 0.2. In each stage, AdamW uses $\beta = (0.9, 0.95)$, weight decay 10^{-2} , peak learning rate 1.128×10^{-3} , 500 warmup steps, cosine decay, and gradient clipping at 1. Headline samples use 32 positions, top- $p = 0.95$, CFG 3.0, and 25 detokenization steps. One RibbonTok “token” is a temporal position of four indices predicted in parallel from one transformer state, so 32 tokens require 32 causal steps, not 128. Reconstruction budgets count encoded positions, whereas generation budgets count sampled positions.

Image Reconstruction

Main comparison. At 256 positions, RibbonTok-XL reaches rFID 1.04, PSNR 19.36, and SSIM 0.582; its 32-position rFID is 1.39. Scaling M to L gives the largest gain (3.82 to 1.27 rFID), followed by a smaller XL improvement. VibeToken-LL reports stronger full-budget reconstruction (rFID 0.40) and broad grid generality, so RibbonTok does not claim the best absolute reconstruction. Its contribution is the conjunction of discrete prefix decodability, explicit common-prefix assignment alignment, and strong 32-position generation. ChannelTok provides an additional contemporaneous flexible-length reference: at 256 channel tokens it reports rFID 3.70, versus RibbonTok-XL’s 1.04, al-

Target	rFID↓	PSNR↑	SSIM↑
256×256	1.04	19.36	0.582
192×192	0.97	20.52	0.616
128×128	0.773	22.31	0.685
64×64	0.613	25.10	0.779

Table 2: Native-grid reconstruction with RibbonTok-XL. Each target is encoded and reconstructed at its model-defined token budget, with metrics computed against references resized to that target. Because lower-resolution rows contain less high-frequency information, rows are not directly comparable.

though its channel-wise token and decoder protocol differ. For the generative detokenizer, rFID measures distributional realism while PSNR/SSIM penalize image-specific completion errors; we therefore interpret the metrics jointly rather than treating low short-prefix rFID as evidence that every detail is already encoded.

Prefix length and resolution consistency. Figure 6 shows monotonic reconstruction improvement as observed-image prefixes add evidence; integration gains largely saturate around 25 steps. Table 2 evaluates direct native-grid reconstruction at four trained square targets. This tests rendering behavior, whereas coalesce matching targets the distinct representation-level property: a decoder may mechanically

Method	Tokens	Latency (s)			Quality	
		AR	Detok.	E2E	gFID↓	IS↑
RibbonTok	32	0.65	2.19	2.87	1.43	348.8
LlamaGen†	576	> 5.51	0.47	> 5.98	2.18	263.3

Table 3: Single-image A100 end-to-end efficiency with 3B AR backbones. RibbonTok is timed end to end. †LlamaGen uses our conservative 256-step AR lower bound plus its reported 0.47-s detokenization; quality is source-reported at 576 tokens.

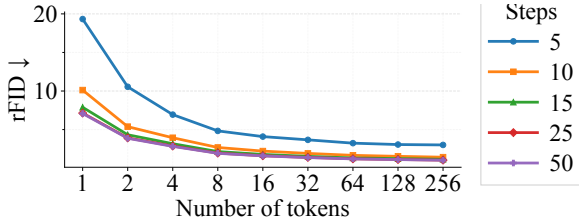


Figure 6: Reconstruction rate-quality sweep. ImageNet-val-256 rFID as prefix length and flow steps vary; gains largely saturate near the 25-step default.

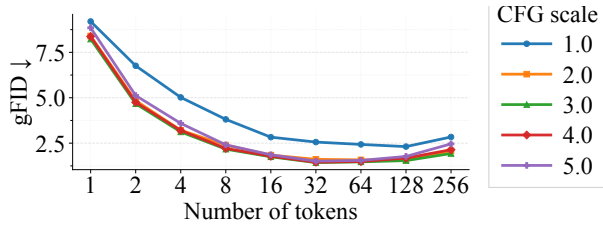


Figure 7: Generation rate-quality sweep. ImageNet-val-256 gFID for the 3B model with 25-step detokenization. Longer generated sequences need not improve because each additional position adds a prediction.

accept several grid sizes without making resized views share discrete identities. At ordinary inference, each image is tokenized independently, so cross-view identity remains learned rather than algebraically guaranteed. The ablation isolates the quality effect of enabling that mechanism, but it is not a direct independent-view token-agreement measurement. We therefore limit the consistency claim to the explicit training objective and the sampled resolution family, not arbitrary unseen resolutions or aspect ratios.

Image Generation

Quality and token budget. With 32 predicted positions, the 1B/3B models obtain gFID 1.68/1.43 and IS 293.7/348.8. The 3B model improves over source-reported LlamaGen-3B (gFID 2.18, IS 263.3) with 32 rather than 576 positions and over VibeToken-Gen’s reported 64-position gFID 3.62. Figure 7 is U-shaped: short prefixes underspecify structure, whereas longer ones accumulate prediction error. Reconstruction instead improves monotonically because its later

Setting	rFID ₃₂ ↓	rFID ₂₅₆ ↓	Aligned
Full model	1.39	1.04	Yes
w/o code normalization	1.63	1.43	Yes
w/o ViT causal mask	1.51	1.03	Yes
w/o trailing-token dropping	15.48	0.92	Yes
w/o stochastic sampling	1.61	1.39	Yes
w/o coalesce matching	1.36	1.01	No

Table 4: RibbonTok-XL design ablations. ImageNet-val-256 rFID uses 32- and 256-position prefixes and 25 detokenization steps. “Aligned” denotes explicit common-prefix coalescing during training, not an inference from same-resolution rFID.

tokens come from the observed image. The best generation range is 32–128 positions; we use 32 for the headline latency-quality point.

End-to-end efficiency. Table 3 shows the compute transfer: 32 AR steps take 0.65 s, 25-step detokenization takes 2.19 s, and the full pipeline takes 2.87 s. Image formation moves from hundreds of causal predictions into spatially parallel flow computation. Our 256-step 3B profile takes 5.51 s, while LlamaGen requires 576 positions; adding its reported 0.47-s detokenization gives the conservative > 5.98-s proxy. RibbonTok is faster and reports better quality, although implementation differences make the comparison approximate. The M/L/XL profiles identify flow as the dominant non-AR cost.

Ablation Study

Table 4 separates the objectives. The causal mask chiefly protects short prefixes: removing it leaves 256-position rFID nearly unchanged but worsens rFID₃₂. Removing trailing-token dropping improves full-budget rFID to 0.92 but degrades rFID₃₂ to 15.48. Normalization and stochastic assignment help both budgets; removing coalescing slightly improves same-resolution rFID but eliminates explicit cross-resolution alignment. The default balances prefix, full-budget, and resolution behavior. Gains diminish near four 16K-entry codebooks; normalized hard-assignment perplexity 0.97 indicates broad marginal use, not uniform tuple occupancy.

Conclusion

We presented RibbonTok, a discrete visual tokenizer that separates a prefix-decodable 1D stream from its rendering grid. Directional codebooks and a rectified-flow DiT support flexible budgets and grids, while coalesce matching trains consistent prefixes across raster changes. RibbonTok-XL reaches rFID 1.39/1.04 at 32/256 positions and 32-position gFID 1.43, with 2.87-s end-to-end latency. Limitations include externally selected budgets, square-resolution training, no direct cross-view identity metric, and 25-step detokenization. Future work should learn content-aware stopping, directly measure cross-view token agreement, broaden training across aspect ratios, and accelerate flow integration.

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